



EAST WEST UNIVERSITY

LAB REPORT 5

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Experiment Title:

To Design, Implement and Performance of a Single Stage BJT Amplifier

Objectives

1. To design a single-stage Common Emitter (CE) amplifier using voltage-divider bias for a specified DC operating (quiescent) point.
2. To implement the designed circuit on a breadboard and experimentally verify its DC bias conditions and small-signal AC performance (voltage gain, phase relationship, and waveform behaviour) using a multimeter/mA-meter and a dual-channel oscilloscope.
3. To know how to operate and measure various properties using oscilloscope
4. To know how a Single Stage BJT Amplifier amplifies signal.
5. To understand the performance and 180-degree phase difference produced by Single Stage BJT Amplifier.

Theory:

A Common Emitter (CE) amplifier is the most widely used BJT amplifier configuration because it provides appreciable voltage gain, current gain and power gain simultaneously. In this configuration the emitter terminal is common to both the input (base) and output (collector) circuits, the input signal is applied at the base and the amplified output is taken from the collector. A distinguishing feature of the CE amplifier is that the output voltage is 180° out of phase with respect to the input voltage, i.e., the amplifier is an inverting amplifier.

The circuit uses a four-resistor voltage-divider bias network (R_1 , R_2 , R_C , R_E), which is the most stable biasing scheme because the operating point is largely independent of transistor β , which varies with temperature and from device to device. The base voltage is fixed by the potential divider formed by R_1 and R_2 , and an emitter resistor R_E provides negative DC feedback that stabilizes the quiescent collector current I_C against variations in β and temperature.

For AC operation, coupling capacitors C_1 and C_2 block the DC bias levels from the signal source and the load respectively while allowing the AC signal to pass. A bypass capacitor C_E is placed across R_E to provide a low-impedance AC path to ground; this removes the negative-feedback action of R_E for AC signals, so the maximum available small-signal voltage gain can be realised. If C_E does not present a sufficiently low reactance at the operating frequency, R_E is only partially bypassed, and the voltage gain is reduced accordingly — this effect is examined quantitatively in the Calculations section of this report.

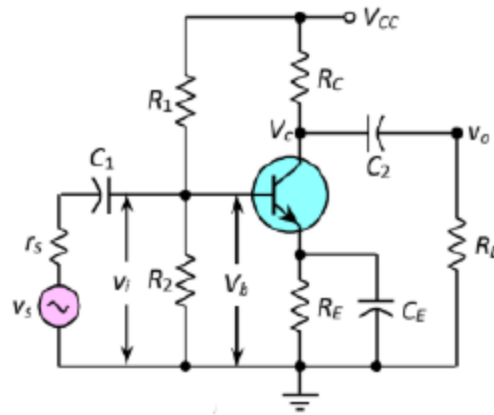


Fig-1 : Single stage CE amplifier with voltage divider bias.

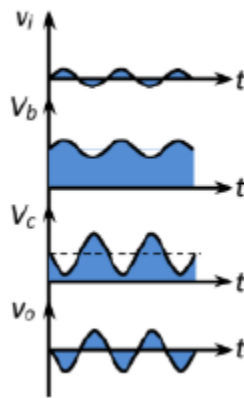


Fig-2 : Voltage waveforms of input source, base terminal, collector terminal and the load.

Components and Apparatus

1. BJT (2N2222)
2. Resistors: 2.2k, 8.2k, 3.3k, 1.5k, 10k
3. Capacitors: 0.1 μ F (3 pieces)
4. Ammeter
5. Oscilloscope
6. Power Supply: DS 12V
7. Breadboard
8. Connecting Wire

Circuit Working Diagram

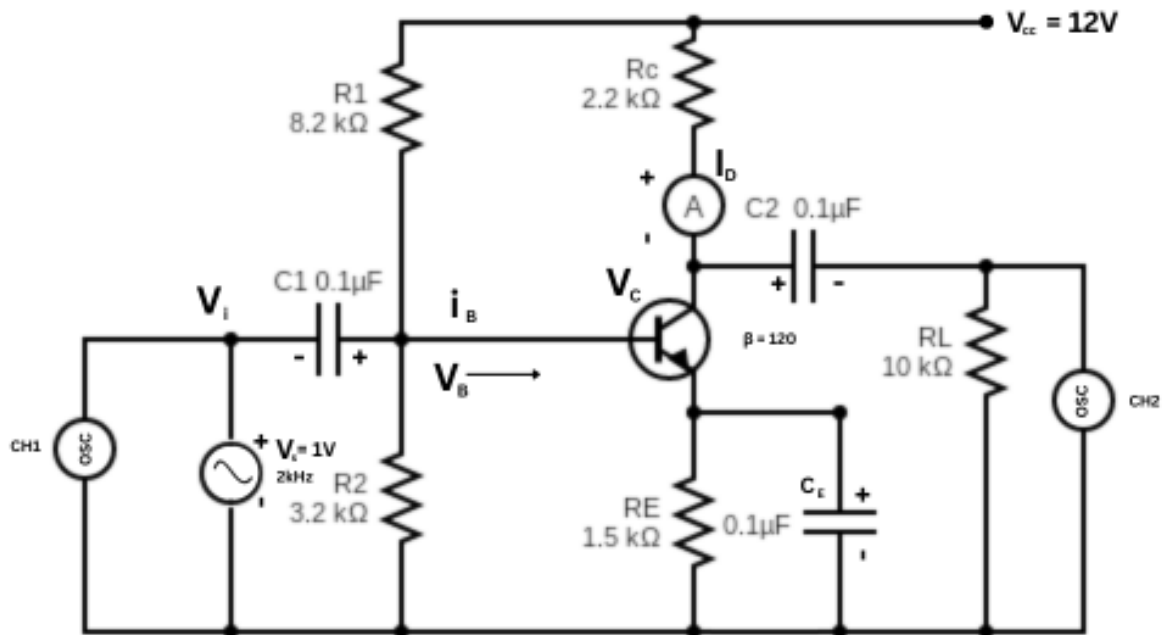


Fig. 1: Designed single-stage CE amplifier with voltage-divider bias

Oscilloscope Readings:

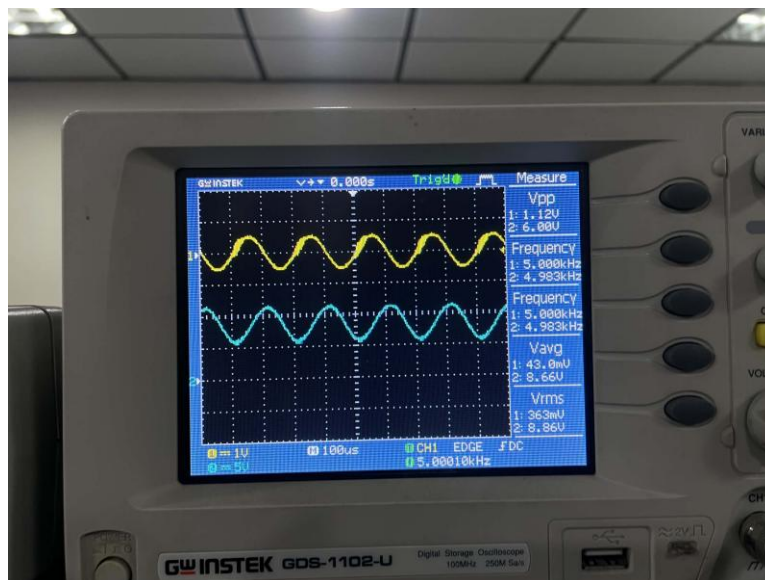


Fig. 4: Trial 1 – Input V_i (CH1, yellow) and output V_o across RL (CH2, cyan). $V_{pp1} = 1.12$ V, $V_{pp2} = 6.00$ V, $f = 5.000$ kHz.

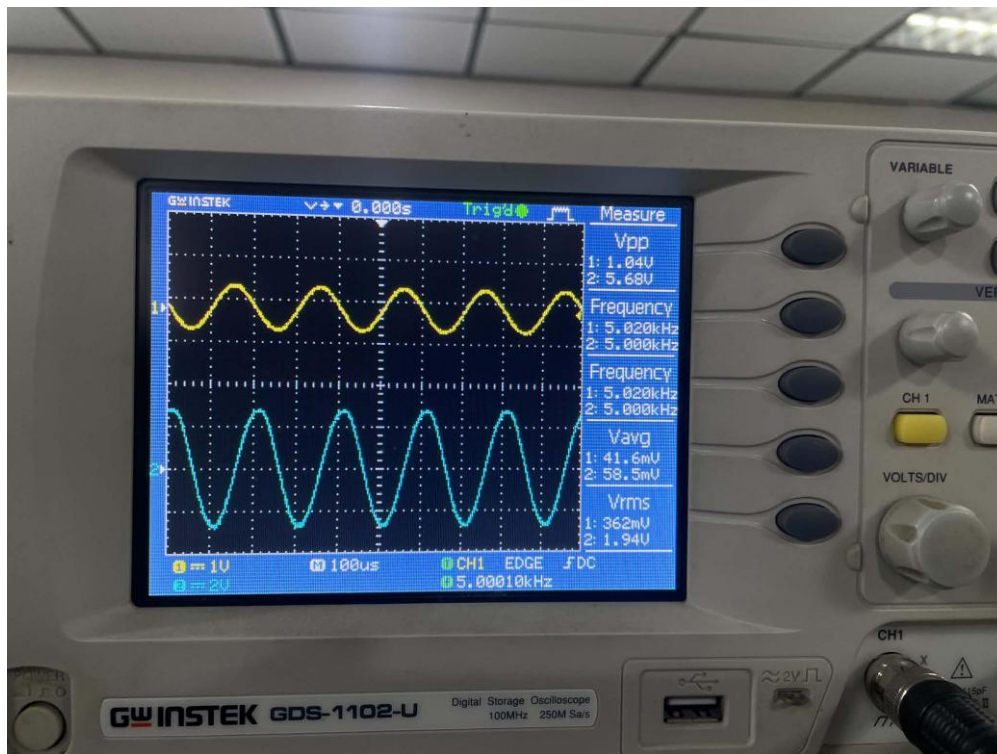


Fig. 5: Trial 2 – Input V_i (CH1, yellow) and output V_o across RL (CH2, cyan). $V_{pp1} = 1.04\text{ V}$, $V_{pp2} = 5.68\text{ V}$, $f \approx 5.0\text{ kHz}$.

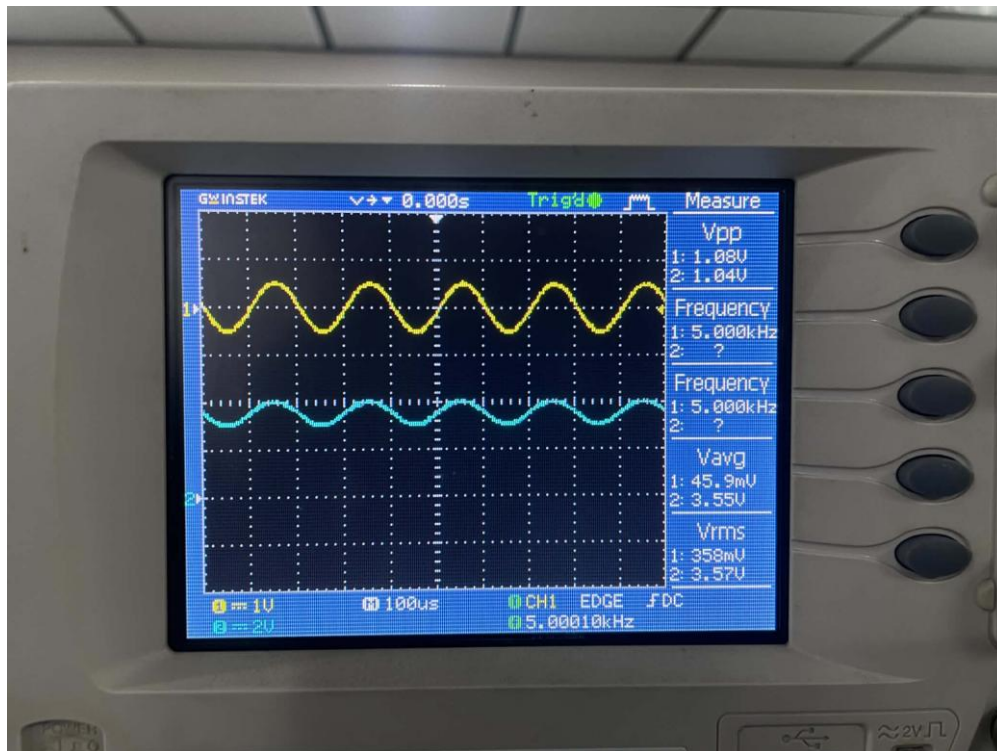


Fig. 6: Input V_i (CH1, yellow) and base voltage V_B (CH2, cyan) showing the small AC signal riding on the DC bias point ($V_{avg2} = 3.55\text{ V} \approx \text{measured } V_B$), confirming correct biasing under signal conditions.

Calculations:

Since the input voltage is provided at the base using a voltage divider network, we use Thevenin equivalent circuit to solve this,

Given,

Supply Voltage, $V_{CC} = 12 \text{ V}$

$R_1 = 8.2 \text{ k}\Omega$

$R_2 = 3.3 \text{ k}\Omega$

$R_C = 2.2 \text{ k}\Omega$

$R_E = 1.5 \text{ k}\Omega$

$\beta = 100$

$V_{BE} = 0.7 \text{ V}$ (Si transistor)

$$\begin{aligned}\text{Thevenin equivalent voltage, } E_{th} &= \frac{V_{CC} \cdot R_2}{R_1 + R_2} \\ &= \frac{12 \text{ V} \cdot 3.3 \text{ k}\Omega}{8.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} = 3.44 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{Thevenin equivalent resistance, } R_{th} &= \frac{R_1 \cdot R_2}{R_1 + R_2} \\ &= \frac{8.2 \text{ k}\Omega \cdot 3.3 \text{ k}\Omega}{8.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} \\ &= 2.35 \text{ k}\Omega\end{aligned}$$

$$\begin{aligned}\text{Base current, } I_B &= \frac{E_{th} - V_{be}}{R_{th} + (\beta + 1)R_E} \\ &= \frac{3.44 \text{ V} - 0.7 \text{ V}}{2.35 \text{ k}\Omega + (100 + 1) 1.5 \text{ k}\Omega} \\ &= 0.0178 \text{ mA} \\ &= 17.80 \mu\text{A}\end{aligned}$$

$$\begin{aligned}\text{Output current, } I_C &= \beta I_B \\ &= 100 \cdot 0.0178 \text{ mA} \\ &= 1.78 \text{ mA}\end{aligned}$$

As I_B is very small, $I_C \simeq I_E = 1.78 \text{ mA}$

$$\begin{aligned}V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 12 \text{ V} - 1.78 \text{ mA}(2.2 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\ &= 5.414 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{Now, } r_e &= 26 \text{ mV}/I_E \\ &= 26 \text{ mV}/1.78 \text{ mA} \\ &= 15.29 \Omega\end{aligned}$$

$$R_C \parallel R_L = \frac{R_C \cdot R_L}{R_C + R_L}$$

$$= \frac{2.2 \text{ k}\Omega * 10 \text{ k}\Omega}{2.2 \text{ k}\Omega + 10 \text{ k}\Omega}$$

$$= 1.8 \text{ k}\Omega$$

$$\text{Theoretical voltage gain, } A_V (\text{theoretical}) = \frac{-R_c || R_L}{r_e} = \frac{-1.8 \text{ k}\Omega}{15.29 \Omega} = -117.72$$

Experimental calculation

Measured, $V_{CE} = 5.87 \text{ V}$

Measured, $I_C = 1.6 \text{ mA}$

Input voltage, $V_i = 1.92 \text{ V}$

Output voltage, $V_o = -4.32 \text{ V}$ (Negative due to 180 degree phase difference)

$$\text{Experimental voltage gain, } A_V (\text{experimental}) = \frac{V_o}{V_i}$$

$$= \frac{-4.32 \text{ V}}{1.92 \text{ V}}$$

$$= -2.25$$

$$\% \text{ of Error in } I_C = \left| \frac{I_C(\text{theoretical}) - I_C(\text{experimental})}{I_C(\text{theoretical})} \right| \times 100$$

$$= \left| \frac{1.78 - 1.6}{1.78} \right| \times 100$$

$$= 10.11\%$$

$$\% \text{ of Error in } V_{CE} = \left| \frac{V_{CE}(\text{theoretical}) - V_{CE}(\text{experimental})}{V_{CE}(\text{theoretical})} \right| \times 100$$

$$= \left| \frac{5.414 - 5.87}{5.414} \right| \times 100 = 8.42\%$$

Results:

After conducting the experiment successfully, we can see,

Measured, $I_C = 1.6 \text{ mA}$

$\% \text{ of Error in } I_C = 10.11\%$

Measured, $V_{CE} = 5.87 \text{ V}$

$\% \text{ of Error in } V_{CE} = 8.42\%$

Theoretical voltage gain, $A_V (\text{theoretical}) = -117.72$

Experimental voltage gain, $A_V (\text{experimental}) = -2.25$

The errors in I_C and V_{CE} are within acceptable margin and so is their value. But we can see a big difference in the theoretical voltage gain and experimental voltage gain. As we cannot measure the maximum gain the experimental result doesn't match the theoretical result.

Discussion

In this experiment, a single-stage Common Emitter BJT amplifier was designed and implemented using a voltage-divider biasing technique. The main purpose was to study the DC operating point of the transistor and observe the amplification and phase relationship of an AC signal. The circuit was built using a 2N2222 transistor and the required resistors, capacitors, power supply, and measurement instruments.

From the theoretical calculations, the collector current, I_C , was found to be 1.78 mA and the collector-emitter voltage, V_{CE} , was 5.414 V. In the practical experiment, I_C was measured as 1.6 mA and V_{CE} was measured as 5.87 V. The percentage errors were 10.11% for I_C and 8.42% for V_{CE} . These differences are acceptable in a practical circuit. The main reasons for the errors can be the variation of the actual transistor β from the assumed value, resistor tolerances, power supply variation, and small losses or loose connections in the breadboard.

The oscilloscope readings showed that the amplifier successfully increased the amplitude of the input signal. In Trial 1, the input voltage was 1.12 Vpp and the output voltage was 6.00 Vpp. In Trial 2, the input voltage was 1.04 Vpp and the output voltage was 5.68 Vpp. These readings confirm that the circuit provided voltage amplification. The output waveform was also observed to be inverted with respect to the input waveform. This confirms the characteristic 180 degree phase difference of a Common Emitter amplifier.

The measured base voltage was approximately 3.55 V, which indicates that the transistor was properly biased around its operating point. The oscilloscope also showed that the AC signal was present around the DC bias voltage at the base terminal. This helped us understand how a small AC signal is applied to the transistor and amplified at the collector.

However, there was a significant difference between the theoretical and experimental voltage gain. The theoretical voltage gain was calculated as -117.72, while the experimental value was -2.25. The negative sign represents the 180 degree phase reversal between the input and output. The large difference can be caused by several practical factors. The emitter bypass capacitor may not completely bypass the emitter resistor at the operating frequency, which reduces the gain. The load resistance also affects the effective collector resistance and therefore reduces the gain. In addition, the actual transistor characteristics, component tolerances, and measurement conditions are different from the ideal assumptions used in theoretical calculations.

Overall, the experiment successfully demonstrated the working principle of a single-stage Common Emitter BJT amplifier. The practical results were reasonably close to the theoretical DC values, and the oscilloscope clearly showed voltage amplification and the expected 180 degree phase reversal. The experiment also provided practical experience in circuit construction, transistor biasing, and using an oscilloscope to measure voltage and observe waveforms.